

Basic Software Requirements Specification

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1 Introduction and Overview

BeAvis is a car rental company that currently is using a pen-and-paper rental process for processing their car. The purpose of this system is to replace the manual process with a software system that makes renting vehicles more efficient and convenient. Embedding this system will smooth the process of renting vehicles which is essential to day-to-day management of the company.

The system should be compatible with the latest version of either IOS or Android. It should also be compatible and easily integrated into websites and applications. Customers will be able to create an account, locate BeAvis locations, search through BeAvis vehicle catalogs, make rentals, sign any legal rental documents, and log their information into their account. Within the company, employees will be able to view customers' requests for rentals, review contracts, view what cars are available for customers, and update the availability of the fleet.

The primary users that this system will affect are the BeAvis customers and employees.

Customers will enjoy a more convenient process for renting vehicles and interacting with the company. Employees will have an easy to manage system that will allow them to manage customer information and cars. This will essentially be a centralized system for employees to access and manage information with customers in an orderly way, unlike the pen-and-paper process.

This Software Requirement Specification (SRS) document is used to describe the requirements of the BeAvis Car Rental System. It's broken down into 3 sections: Customer's and Employee Perspective, Functional system requirements, additional assumptions and constraints for this project. Section 2 will be used to describe the requirements from the employee and the customer perspective. Section 3 will highlight system requirements and use cases of this system in detail. Section 4 will show possible obstacles and necessity for the development of this system.

2 User Requirements

The BeAvis Car Rental System will mainly be used by 2 users, employees and customers. These two types of users will interact with this system in a different way.

Customers will be able to access this system using either the BeAvis mobile application or the BeAvis website. A customer will need to be able to create an account within the system to register their user onto the client-facing application. The user will need to create a username,

password, email, and phone number to create their account. For first time users, they will be prompted to set up their account and provide basic identification information, alongside signing rental car agreements. The customer will also need to allow access to their phone's location to pinpoint the closest BeAvis location. A customer who has already registered will just need to log in using their login credentials that they used during the registration phase. Once logged in, customers will be able to search for BeAvis locations nearby, view what vehicles are at that location, make a rental request to the location, and view their personal information.

The customer will also have to access their account information and settings. Within the setting, they are able to change any accessibility settings, for example, brightness and contrast. This also will be a place for customers to add payment methods and optionally enable two-step authentication using their personal information, or an authenticator app. (Assumption)

This system gives the customer the ability to electronically receive and sign the rental agreements. These agreements will be tied to their account and will have to be stored in a securely locked system. This system will also need to store payment information therefore it would be another secure position to store card information. (Assumption)

Employees will have their own method of logging into the system. Once the employees have been authenticated into the system, they will be able to access the catalog of customer information and vehicles of the location. Employees will be able to view basic information of the customer to ensure privacy and security. They will also be able to determine which vehicles are available to rent, update the status of the vehicle information at the location, and support customers through a chatbox. Employees will primarily use the website or the BeAvis personal software system to manage customer and fleet information. (Assumption: BeAvis having a personal system)

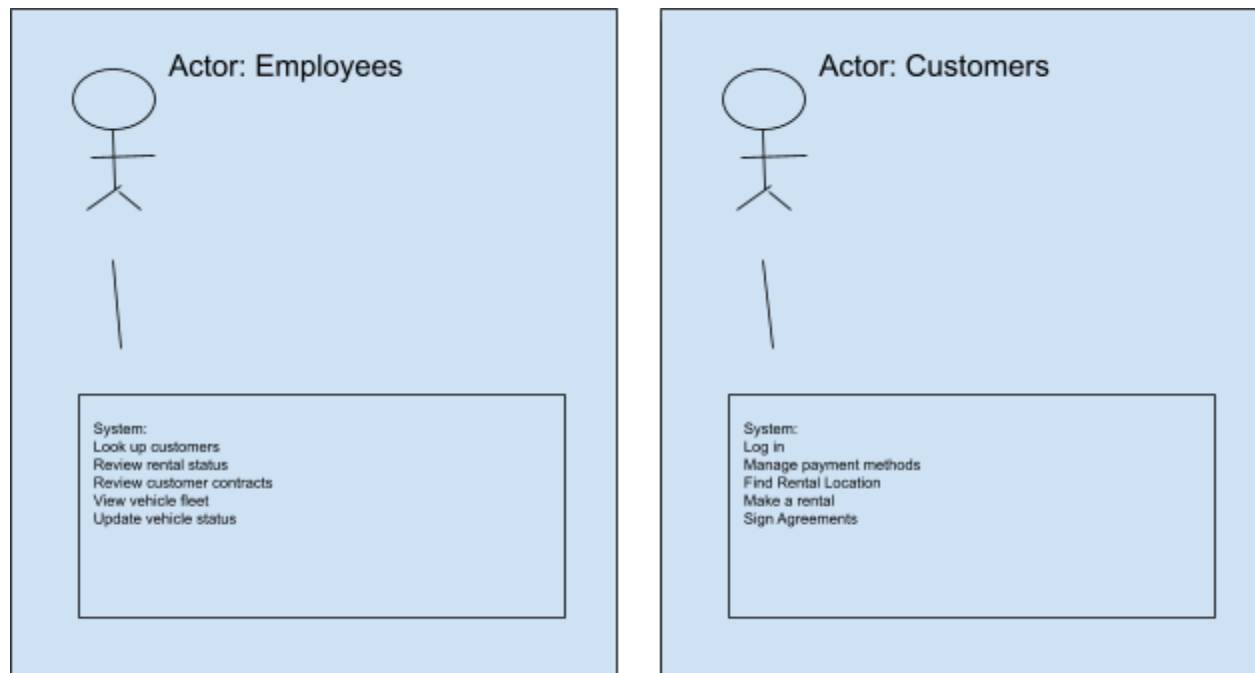
The mobile application must be compatible with the latest version of IOS or Android. The website will provide the encapsulating information about the system alongside the application. The app must need to provide necessary information to complete the rental function or customers, although the appearance of both platforms may differ.

3 System Requirements

3.1 Functional Requirements

The functional requirements describe the services and functionality of the BeAvis Car Rental System. The two primary avatars that will be using this system are Customers and Employees.

Use Case Diagram



The Employee Actor is assigned to look up customer information, review rental status, review customer contracts, view vehicle fleets, and update vehicle status.

The Customer Actor is assigned to log into their account, manage payment methods, find rental locations, make a rental, and sign agreements for renting the car.

Employee Use cases

3.1.1 Look up customers

An employee can search for and access customer information in order to certify the customer picking up the car. This can be a security measure to ensure the person renting is the same person picking it up

3.1.2 Review Rental Status

An employee can view the current status of the customer's rental after the customer has picked up the car. They can also view the day of pickup and customer information according to that car.

3.1.3 Review Customer Contract

An employee can access and review that the customer has signed correctly.

3.1.4 View Vehicle Fleet

An employee can view what vehicles are available and view the company's vehicle fleet capacity.

3.1.5 Update Vehicle Status

An employee can update the status of a vehicle. Vehicles that require maintenance can be removed from the pool of availability, so customers don't get un-functional cars.

Customer Use Case

3.1.6 Log In

The customer enters their account credential to gain access into the BeAvis System. The system will have to check that the credential the customer enters in is correct, if correct the system allows the customer to view information about their profile and fleet availability.

3.1.7 Manage Payment Methods

The customer can add, change, or manage their payment method through the account setting. Customers can use either card, direct bank connection, or other forms of payment. Payment information must be securely stored and kept from the customer's basic information.

3.1.8 Find Rental Location

The customer can use their location services to find nearby BeAvis rental locations. The system will require the user to accept the location request notification to provide accurate nearby location results.

3.1.9 Make a Rental

The customer can request for a rent on a vehicle from a BeAvis location. The system will give the customers car options that are at that current BeAvis location and will provide directions to the locations, as well as procedure to rent.

3.1.10 Sign Agreements

The customer will need to receive and electronically sign the rental agreement. The system will then securely store the agreements and customer signature.

3.2 Non-functional Requirements

3.2.1 Security

The system should protect both the customer and employee information from any unauthorized access. Users must have a way to authenticate and verify that they are the one using the system, before interacting with the system. Customer payment information must be securely locked and separated from customer basic information for rentals. This separation of system will reduce the ability for customer sensitive information to be leaked to other users. Along with payment information, contracts must be kept organized to the user and securely locked. Two-factor authentication is highly recommended to users and employees.

3.2.2 Privacy

The system has to be able to protect sensitive information, such as customer address, signatures, account information, and rental information. Accessing this information should be locked to that specific user.

3.2.3 Usability

The system's website and mobile application should provide a simple and intuitive design to make sure users can easily locate the main functions. The website and application do not have the same design. The goal for each actor should be the forefront of the design architecture.

3.2.4 Compatibility

The BeAvis system should be accessible through both the website and mobile application. The mobile application should be able to support the latest version of IOS and android. The website must have the same functionality as the application to be able to complete a car rental.

3.2.5 Location Services

The BeAvis system must request the location permission to the customer before accessing the customer information. When permission has been granted, the location should be used to identify any BeAvis Car Rental locations and be able to provide directions to the location.

4 Other

4.1 System Constraints

The BeAvis system must provide a website and a mobile application. The mobile application needs to be able to use the latest version of either IOS or androids.

Customers must provide accurate and valid information about the customer and be able to complete an email verification before their account can be used. Two-step verification can be optional and cannot be turned on by default.

4.2 Assumptions

Some details necessary for defining the system are not explicitly stated in the provided BeAvis description. The assignment permits reasonable assumptions as long as important assumptions are identified. The following assumptions are therefore used:

- Customer and employee accounts have different access permissions.
- Only authenticated employees can modify vehicle availability or maintenance status.
- A vehicle marked as requiring maintenance cannot be rented by a customer.
- Each rental is associated with a customer and a vehicle.
- Customers must complete the required rental agreement before a rental is finalized.
- BeAvis maintains information about its locations and vehicles within the system.
- Customer payment information is not directly visible to employees who look up customer rental information.